

List Manipulation

filter-atom

Description: Filters a list of atoms based on a predicate.

Parameters:

- List: The list of atoms to filter.
- Variable: The variable to use in the predicate.
- Filter: The predicate to apply (an expression that returns `True` or `False`).

Return: A new list containing only the atoms that satisfy the predicate.

Example:

```
!(filter-atom (1 2 3 4 5) $x (> $x 3)) ; Returns (4 5)
```

map-atom

Description: Applies a function to each atom in a list, creating a new list with the results.

Parameters:

- List: The list of atoms to map over.
- Variable: The variable to use in the mapping expression.
- Map: The expression to apply to each atom.

Return: A new list with the mapped values.

Example:

```
!(map-atom (1 2 3) $x (+ $x 1)) ; Returns (2 3 4)
```

foldl-atom

Description: Folds (reduces) a list of values into a single value, using a binary operation. This is a left fold.

Parameters:

- List: The list of values to fold.
- Init: The initial value.
- A: The variable to hold the accumulated value.
- B: The variable to hold the current element of the list.
- Op: The binary operation to apply (an expression using `A` and `B`).

Return: The final accumulated value.

Example:

```
!(foldl-atom (1 2 3 4) 0 $acc $x (+ $acc $x)) ; Returns 10 (1+2+3+4)
```

format-args

Description: Replace the '{}' in the expression (the first argument) with an atom (the second atom).

Parameters:

- ExpressionAtom: Expression with {} to be replaced with the second parameter.
- Atom: Atom to replace {} in the expression.

Return: Expression with replaced {} with atoms

Example:

```
!(format-args (Probability of {} is {}%) (head 50)) ; Returns (Probability of head is 50%)
```